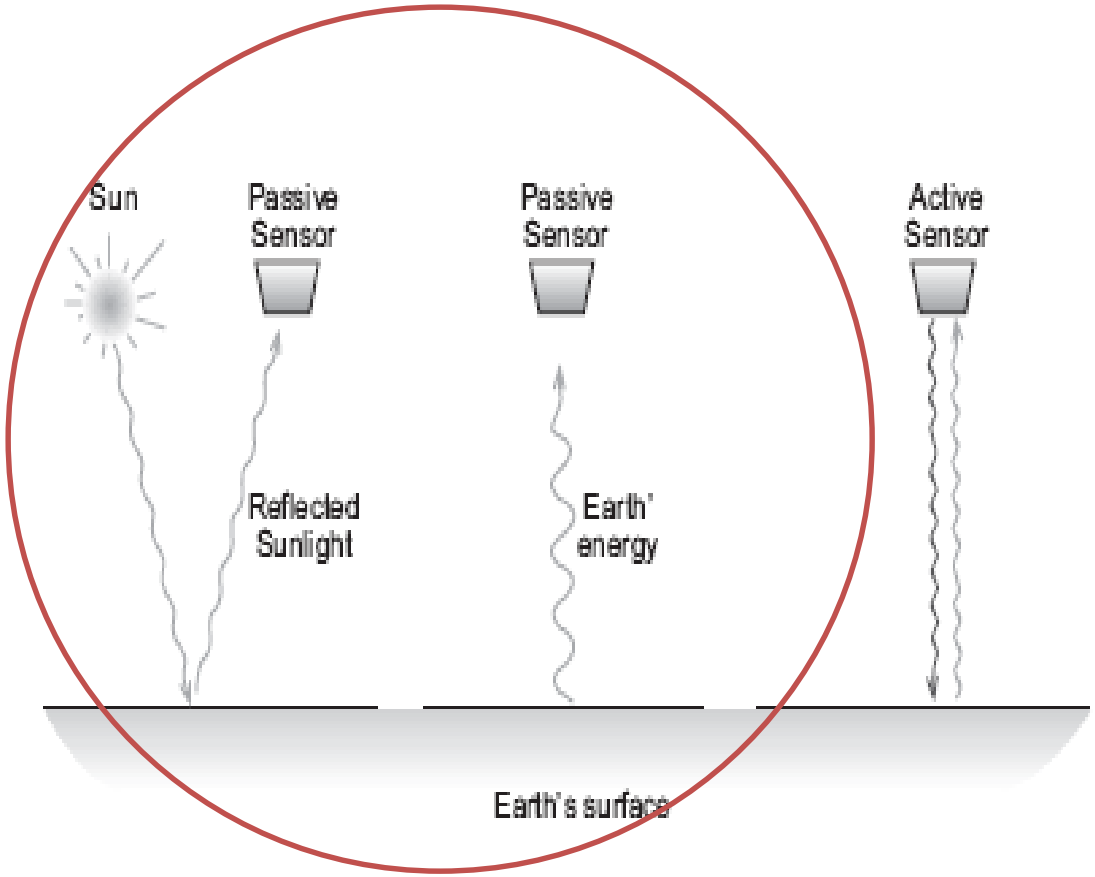

بسم الله الرحمن الرحیم

Electromagnetic Radiations, Spectrum and Energy Interaction Remote Sensing Perspective

Introduction

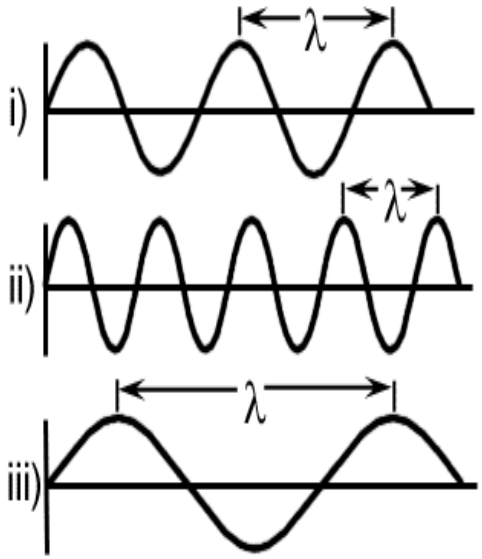
- Remote sensing relies on the measurement of electromagnetic (EM) energy
- Interpretation of remote sensing imagery depends on a sound understanding of electromagnetic radiation and its interaction with the atmosphere and the surfaces
- EM energy can take several different forms
- The most important source of EM energy at the Earth's surface is the Sun

Sensors


Sun
Passive Sensor
Passive Sensor
Active Sensor
Reflected Sunlight
Earth' energy
Earth's surface

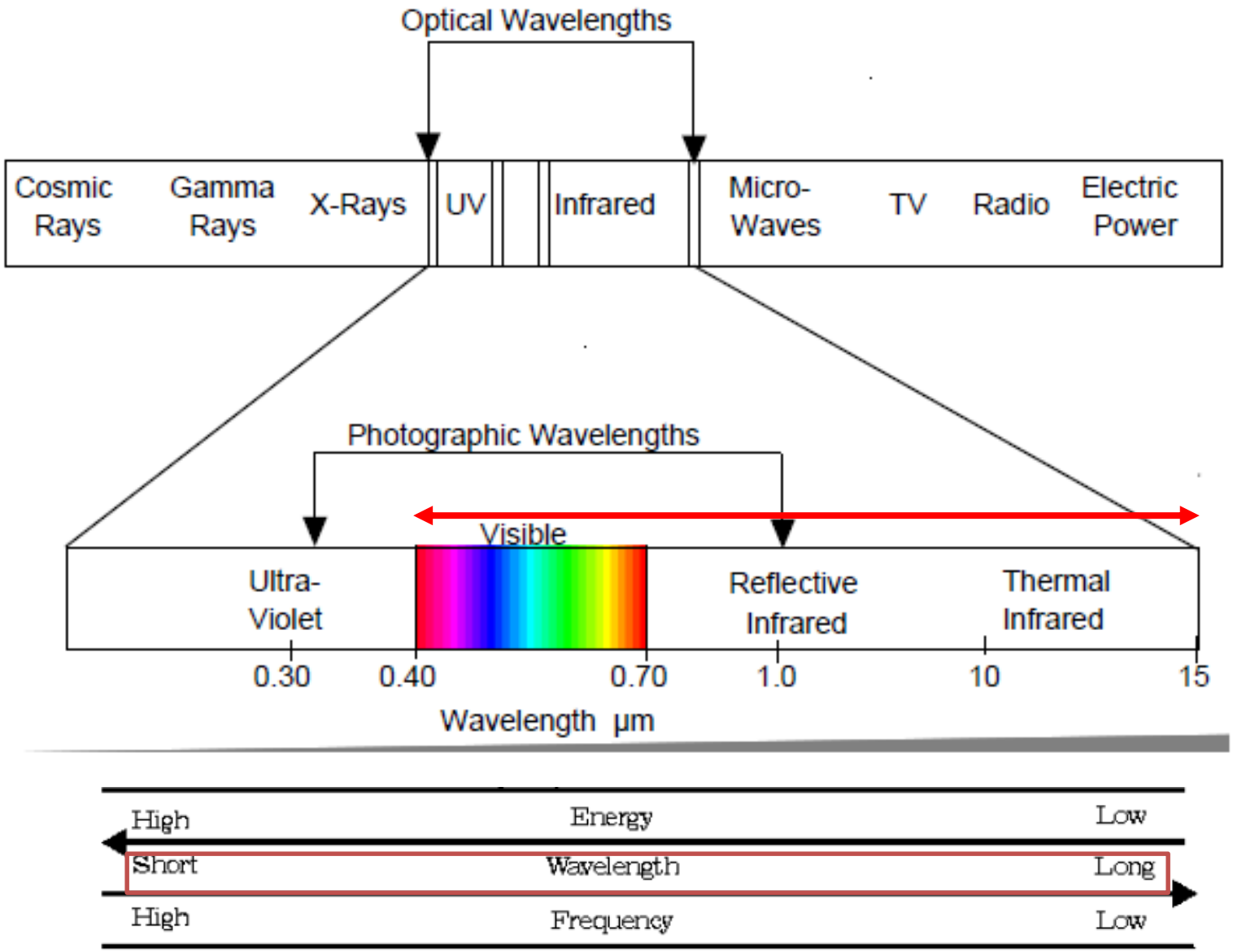
Important Terms

- **Wavelength (λ)**
  - Units: Micrometer or Nanometer

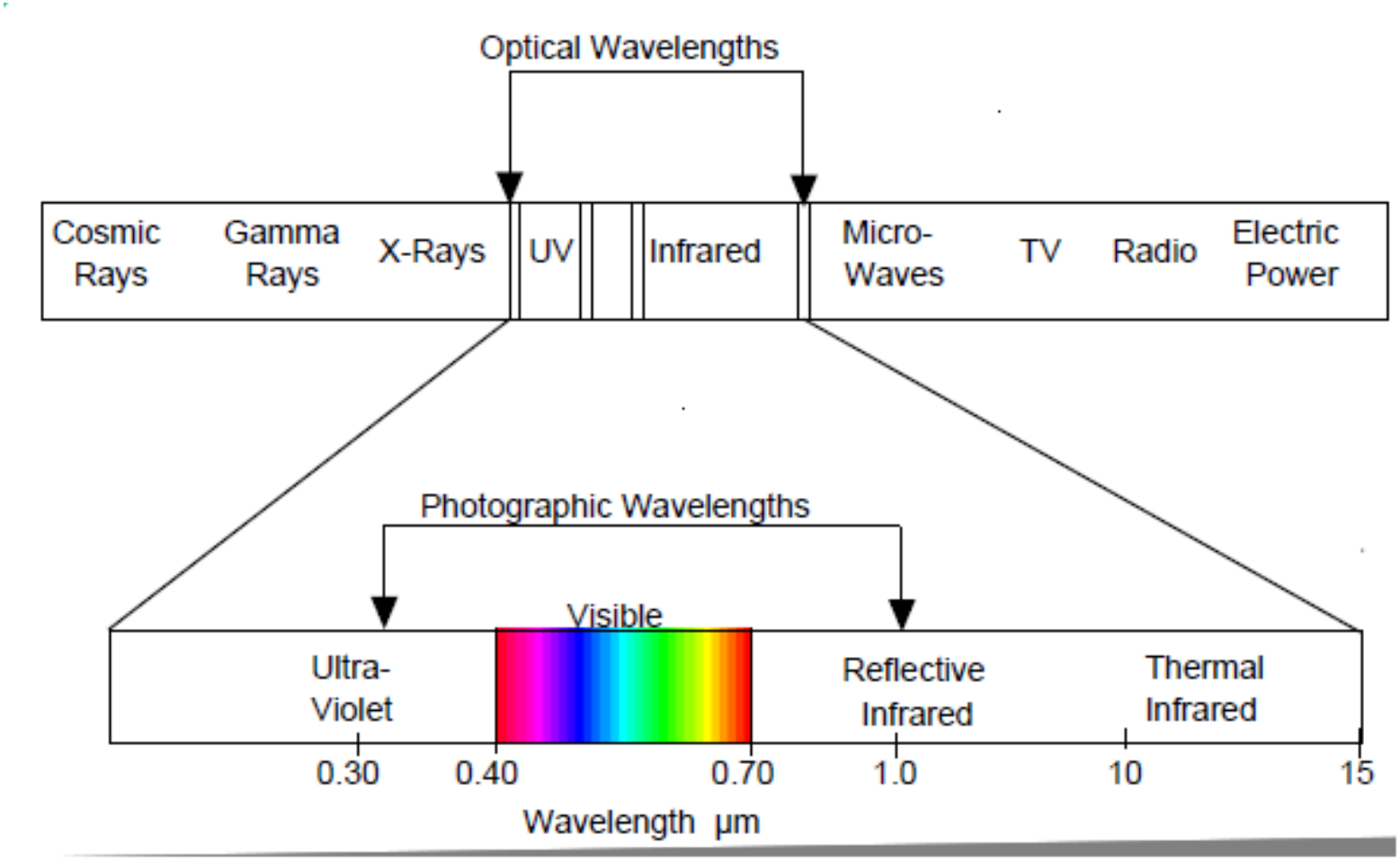


Electromagnetic Spectrum

- Electromagnetic Spectrum ranges from shorter wavelengths (including gamma and x-rays) to longer wavelengths (including microwaves and broadcast radio waves)
- Of the whole electromagnetic spectrum, a few regions are useful for remote sensing (Visible to IR)


Optical Wavelengths
Cosmic Rays
Gamma Rays
X-Rays
UV
Infrared
Micro-Waves
TV
Radio
Electric Power
Photographic Wavelengths
Visible
Ultra-Violet
Reflective Infrared
Thermal Infrared
0.30
0.40
0.70
1.0
10
15
Wavelength µm

High
Energy
Low
Short
Wavelength
Long
High
Frequency
Low



- **EM spectrum is arbitrarily segmented into major divisions**
- **There are no natural breaks in the EM spectrum**
- **These separations are made for our convenience**

Electromagnetic Spectrum

- **The light that our eyes- and "remote sensors" - can detect is part of the *Visible Spectrum.***
- **The visible portion is much smaller relative to the rest of the spectrum**
  - **Wavelength range from approximately 0.38 µm to 0.7 µm**
- **Many radiations around us are "invisible" to our eyes but can be detected by other remote sensing instruments and used to our advantage**

Electromagnetic Spectrum

- ***Infrared (IR)*** **region wavelength ranges from approximately 0.7 µm to 14/15 µm**
- **Infrared regions can be divided into two categories based on their radiation properties.**
  - **Reflected IR (Near IR)- essentially solar radiation reflected from Earth's surface (0.7 µm to 3.0 µm)**
  - **Emitted or Thermal IR (Far IR)- radiations emitted by the Earth, heat and thermal energy (3.0 µm to 14.0 µm)**

Electromagnetic Spectrum

- Radiations in the reflected IR region are used for remote sensing purposes in ways similar to those of radiation in the visible portion.

- Thermal IR region is quite different from the visible and reflected IR regions, as this energy is essentially the radiation that is emitted from the Earth's surface in the form of heat.

Interaction with the Atmosphere

- Atmospheric modification of incoming and outgoing

  EM radiation includes:

  - Scattering
  - Refraction
  - Absorption

Interactions With The Atmosphere

Scattering

- Radiations scattered within the atmosphere and which reach the sensor without contact with the Earth's surface are called atmospheric path radiance.
- Occurs when particles or large gas molecules present in the atmosphere interact with and cause the electromagnetic radiation to be redirected from its original path

Interactions With The Atmosphere

- **Scattering depends on several factors, including**
  - The wavelength of the radiation
  - Size and abundance of suspended particles or gases
  - The distance the radiation travels through the atmosphere

Interactions With The Atmosphere

- **Rayleigh Scattering or clear atmosphere scattering**
  - **Particle size very small, mostly effect Shorter WLs**
- **Mie Scattering**
  - **Particle size about equal, Longer WLs**
- **Nonselective Scattering**
  - **All wavelengths are scattered about equally**
  - **Particle size much larger**

Interactions With The Atmosphere

- Sky appears “blue” during the day because of Rayleigh Scattering
- As sunlight passes through the atmosphere, the shorter wavelengths (i.e., blue) of the visible spectrum are scattered more than the other (longer) visible wavelengths

Interactions With The Atmosphere

- At sunrise and sunset, the light has to travel farther through the atmosphere than at midday
- Thus, the scattering of the longer wavelengths, a greater proportion of these can penetrate the atmosphere

General Effects Of Scattering

- Causes skylight scattering, which allows us to see items within a shadow
- Forces image to record the brightness of the atmosphere in addition to the target
- Reduces contrast,
  - decrease in the detectability of features present in the image

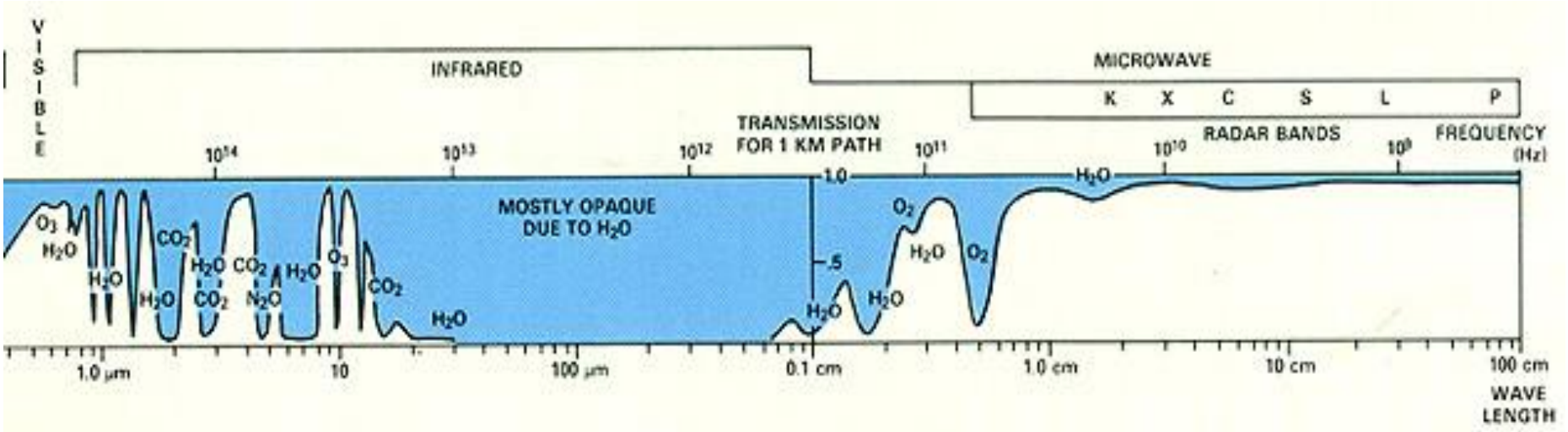
Interactions with the Atmosphere

Absorption

- Causes molecules in the atmosphere to absorb energy at various wavelengths
- Ozone, carbon dioxide and water vapour are the three main atmospheric constituents which absorb radiation
- Because these gases absorb electromagnetic energy in very specific regions of the spectrum, they influence where (in the spectrum) we can “look” for remote sensing purpose

Atmospheric Transmission

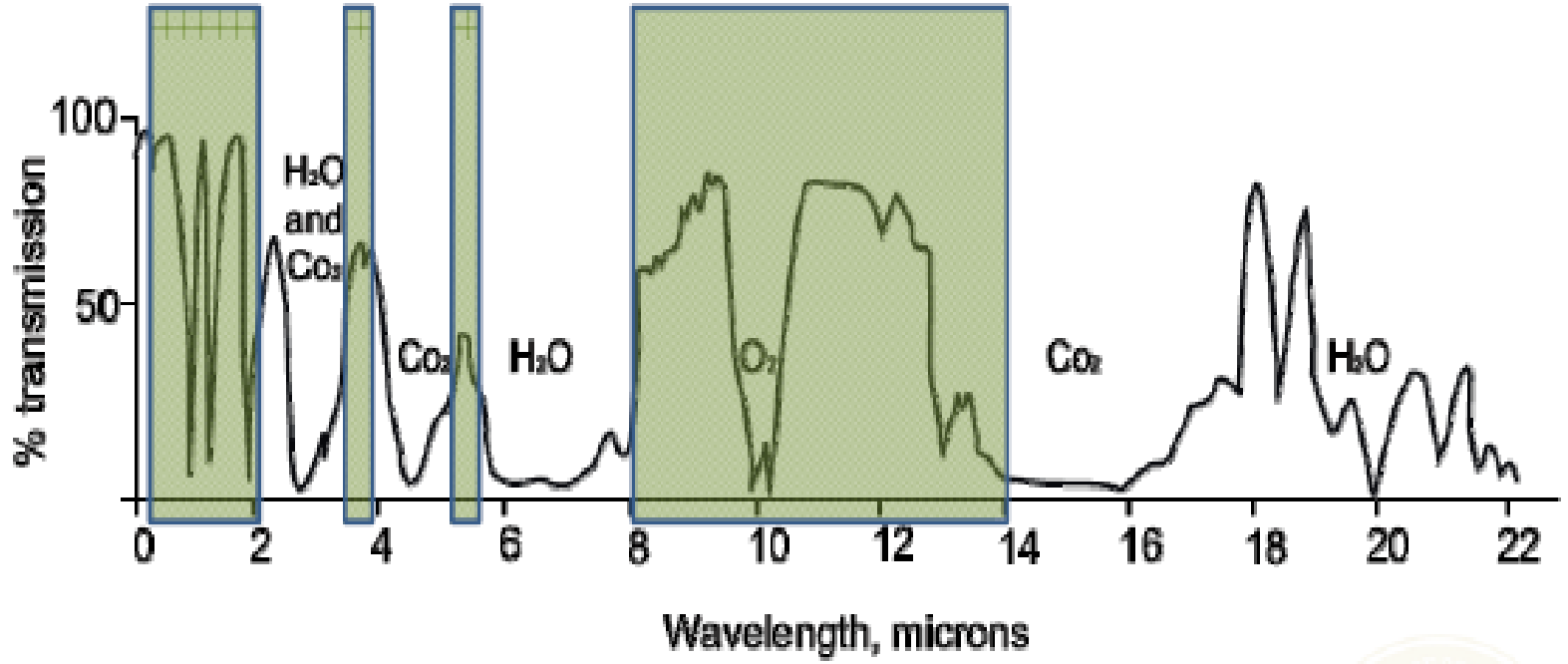
Generalized diagram showing relative atmospheric transmission of radiation of different wavelengths



- Blue zones mark minimal passage of incoming and/or outgoing radiation
- whereas white areas denote "atmospheric windows" in which the radiation experiences much reduced interactions and hence can penetrate the air with little or no loss by absorption

Atmospheric Windows

Regions of the spectrum that are relatively (but not completely) free from the effects of scattering and absorption



- **Window between 0.4-2 μm where the (optical) remote sensors operate as well as the human eye**
- **Windows in the thermal infrared region, around 3 and 5 μm,**
- **Relatively broad, window extending from approximately 8 to 14 μm**

Scattering and Absorption

- Both scattering and absorption limit the usefulness of some portions of the electromagnetic spectrum for remote sensing purposes
- The effect of scattering and absorption
  - Add a degree of haze to the image that is, to reduce the contrast of the image.
  - Reduce the amount of radiation returning to the sensor from the Earth's surface, known as attenuation or extinction (weakening)

Interaction with Earth Features

- Radiations that are not absorbed or scattered in the atmosphere reach and interact with the Earth's surface.
- Interaction with the target depends upon the properties of both the target and the radiation.
- Three things happen to the sun's energy as it reaches the Earth:
  - Reflected
  - Absorbed
  - Transmitted
- The proportions of each will depend on the wavelength of the energy, the material, and condition of the feature.

Reflection

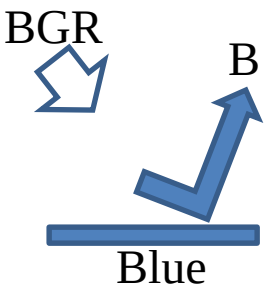
Specular or mirror-like reflection:

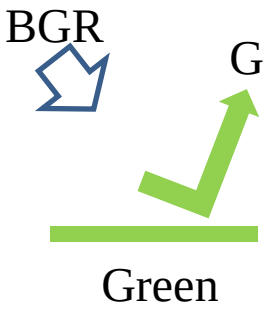
- When a surface is smooth, almost all the energy is directed away from the surface in a single direction
- Specular reflection can be caused, for example, by a water surface or a glasshouse roof.
- It results in a very bright spot in the image.

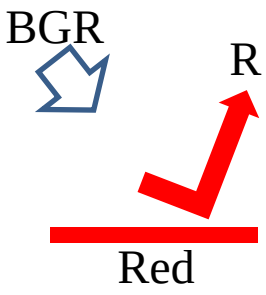
Reflection

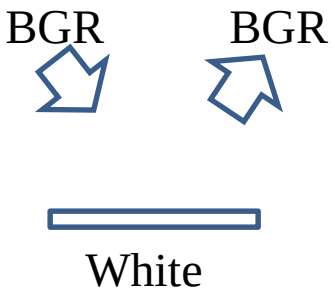
- **Diffuse reflection**
  - **When the surface is rough, and the energy is reflected almost uniformly in all directions**

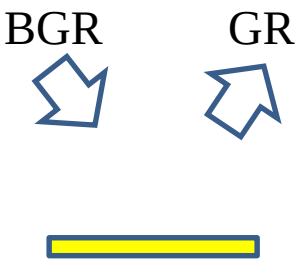
Color of an object is defined by the color of the light it reflects

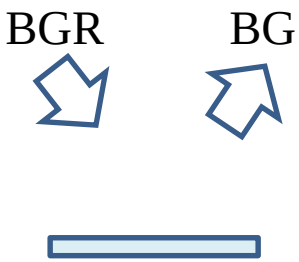


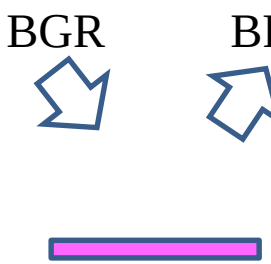














Recording of Energy By The Sensor

- After the energy has been reflected, scattered by, or emitted from the target, a sensor collects and records it

Reference/ Reading

- Computer Processing of Remotely Sensed Images, An Introduction, 3rd edition, Paul, M Mather
  - Chapter 1 Sec 1.1 to 1.3.1

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